

Master Flight Controller - Installation Guide

Production-Ready Autonomous Trading System

SYSTEM REQUIREMENTS

Hardware (Mac Studio M2)

- Mac Studio M2 Max/Ultra
- 32GB+ RAM minimum (64GB recommended)
- 1TB+ SSD storage
- Dual monitor setup (trading + monitoring)
- UPS backup power (2-hour minimum)
- Dedicated ethernet connection (no WiFi)

Network Infrastructure

- Primary ISP: Business-grade fiber (100Mbps+)
- Backup ISP: 4G/5G mobile hotspot (auto-failover)
- VPN to exchange regions (Tokyo, London, New York)
- Target latency: <200ms to primary exchanges

Exchange Access

- API keys for all target exchanges (read + trade permissions)
 - WebSocket access enabled
 - Rate limits documented per exchange
 - Separate keys for paper trading
-

PHASE 1: DATA INFRASTRUCTURE

1.1 Market Data Feeds

Primary Sources:

- Exchange direct WebSocket (BTC, ETH, major pairs)
- Backup REST API polling (200ms interval)

- Order book depth (Level 2 minimum)

Secondary Sources:

- Historical OHLCV data (2 years minimum)
- Tick data for slippage modeling (last 6 months)
- Funding rate data for perpetual futures

Storage Structure:

```
/data
/live      # Real-time streaming cache (last 24h)
/historical # Compressed archives (2+ years)
/features   # Precomputed Fibonacci levels, indicators
/models     # Trained CNN/RNN weights
/states     # HMM parameters, KNN database
```

1.2 Latency Measurement System

- Ping monitoring daemon (every 15 minutes)
 - Log: timestamp, exchange, latency_ms, packet_loss
 - Store P99 latency per exchange
 - Alert if P99 > 200ms for 1 hour
-

PHASE 2: LAYER INSTALLATION

Layer 0: Order Flow Monitoring

Components:

- Bid-Ask Volume Delta calculator
- Real-time order book imbalance tracker
- Absorption detection (price vs volume divergence)

Outputs:

- Volume delta: normalized [-1, 1]
- Imbalance ratio: $\text{buy_volume} / (\text{buy_volume} + \text{sell_volume})$
- Update frequency: every tick

Layer 1: Fibonacci Dynamic Scaler

Components:

- Rolling high/low detector (configurable lookback: 24h, 7d, 30d)
- Fibonacci level calculator (0, 0.236, 0.382, 0.5, 0.618, 0.786, 1.0)
- Multi-symbol normalization engine

Configuration:

- Default lookback: 24 hours
- Recalculation: every 15 minutes
- Anchor point strategy: swing high/low detection

Outputs:

- Normalized price position [0-1] per symbol
- Current Fibonacci zone identifier
- Zone strength score

Layer 2: Neural Avionics

CNN (Pattern Recognition):

- Input: Last 15 minutes of normalized OHLCV (5x symbols)
- Architecture: 3-layer convolutional network
- Output: 128-dimensional pattern feature vector
- Inference device: Apple Neural Engine (ANE)
- Update frequency: every 1 minute

RNN (Temporal Memory):

- Input: Last 60 minutes of CNN features
- Architecture: LSTM with 2 hidden layers
- Output: 128-dimensional trend feature vector
- Inference device: Apple Neural Engine (ANE)
- Update frequency: every 1 minute

Combined Output:

- 256-dimensional latent state vector
- Compression: store last 24 hours only
- Normalization: sigmoid activation for scoring

Layer 3: The Dispatcher

HMM Regime Detector:

- States: Trend, Range
- Observable features: ATR, ADX, volume, return autocorrelation
- Update: state probabilities every 15 minutes
- Retraining: Baum-Welch algorithm every 6 hours

Pareto Risk Filter:

- Calculate tail index for all symbols (last 500 data points)
- Rank symbols by heavy-tail score
- Select top 20% (vital few)
- Update: every 5 minutes

Correlation Matrix Filter:

- Rolling 30-minute correlation across selected symbols
- Drop symbols with correlation >0.85 to existing positions
- Keep highest Sharpe ratio in correlated clusters
- Update: every 1 minute

KNN Strategy Selector:

- Database: historical successful trades with state vectors
- K-value: 15 neighbors
- Distance metric: cosine similarity
- Output: recommended model weights for current state

Layer 4: ILS Precision Guidance

Linear Regression Components:

- Localizer: rolling midline (last 60 data points)
- Glide slope: regression slope and R^2
- Tracking error: price deviation from beam
- Acceleration guard: 2nd derivative of slope

Update Frequency:

- Refit regression: every 15 minutes
- Calculate d^2/dt^2 : every 1 minute
- Quality threshold: $R^2 > 0.85$ for "stable beam"

State Outputs:

- On-Beam: price within $\pm 2\sigma$ of regression
- Off-Beam: price $> 2\sigma$ deviation
- Glide Slope Unstable: acceleration reversal detected

Layer 5: Fuzzy Logic Captain

Confidence Score Calculation:

Confidence = $0.25 \times \text{CNN_signal (sigmoid normalized)}$
+ $0.25 \times \text{RNN_signal (sigmoid normalized)}$
+ $0.20 \times \text{ILS_quality (R}^2 \text{ value)}$
+ $0.15 \times \text{Pareto_score (min-max normalized)}$
+ $0.10 \times \text{HMM_regime (state probability)}$
+ $0.05 \times \text{OrderFlow_delta (tanh normalized, shifted [0,1])}$

State Zones:

Confidence	State	Position Size	Order Type
0.85-1.00	EXECUTE	Full Kelly	Aggressive limit (1 tick inside)
0.65-0.85	SCALE_IN	50% Kelly	Mid-price limit
0.45-0.65	HOLD	No change	Monitor only
0.30-0.45	SCALE_OUT	-50% position	TWAP over 60s
0.00-0.30	ABORT	Close all	Market order

Hysteresis:

- Require 5-minute confirmation before state transitions
- Threshold crossing: must exceed by 0.05 to prevent oscillation

Friction Layer (Pre-Execution Filter)

Slippage Model:

Slippage = base_spread
+ (order_size / L2_depth) × volatility_5min
+ (2 × P99_latency / 1000) × price_velocity

Effective Edge Calculation:

Effective_Edge = Confidence_Score - (Spread + Slippage + Fees)
Execute only if Effective_Edge > 0.15

Parameters to calibrate:

- L2_depth: measure from order book every 6 hours
 - P99_latency: from latency monitoring system
 - Fees: exchange-specific (maker/taker tiers)
-

PHASE 3: RISK MANAGEMENT SYSTEM

Position Sizing (Pareto-Kelly)

Formula:

Kelly_Fraction = (Win_Rate × (1 + Avg_Win/Avg_Loss) - 1) / (Avg_Win/Avg_Loss)
Position_Size = Kelly_Fraction × Available_Capital × Confidence_Score

Inputs:

- Win_Rate: from last 50 trades of specific symbol+model combo
- Avg_Win/Avg_Loss: from same 50-trade window
- Available_Capital: after reserving 50% cash

Hard Limits:

- Max 30% capital per symbol
- Max 5 concurrent positions
- Max 50% total capital deployed
- Max 0.25 Kelly fraction (quarter-Kelly safety cap)

Circuit Breakers

Level 1 (5% drawdown in 60 minutes):

- Halt all new entries for 2 hours
- Maintain existing positions with tightened stops (0.5× normal)
- Log incident for post-analysis

Level 2 (10% drawdown in 24 hours):

- Reduce all position sizes to 50%
- Halt system until manual review
- Require equity recovery above 95% of peak before resuming

Level 3 (Flash Crash - 5% move in 30 seconds):

- Immediate halt of all new entries
- Tighten stops to 0.25× normal
- Switch to manual mode for affected symbols

Tail Risk Hedging

- Allocate 2-5% of capital to VIX options or inverse correlation ETFs
- Rebalance weekly
- Monitor portfolio beta daily

PHASE 4: EXECUTION ENGINE

Order Routing Logic

By Confidence Zone:

- **>0.85:** Aggressive limit → market after 2s if unfilled
- **0.65-0.85:** Passive limit at mid-price → cancel after 10s if unfilled
- **Exits <0.30:** Market order immediately

Order Types:

- Limit orders: all entries
- Market orders: emergency exits, abort states
- TWAP: strategic exits (confidence 0.30-0.45)

Exit Logic (Parallel Monitoring)

Trigger Conditions:

Trigger	Action	Method
Confidence drops >0.20 in 5min	Close 50%	TWAP 30s
Confidence crosses <0.45	Full exit	TWAP 60s
ILS beam break + confidence <0.60	Emergency exit	Market order
Stop loss hit	Close all	Market order

Stop Loss Placement:

- Normal mode: 2× ATR below entry
- Turbulent mode (low ILS R²): 1× ATR below entry
- Glide slope unstable: 0.5× ATR below entry

PHASE 5: DEGRADED MODE OPERATIONS

VFR Mode (Visual Flight Rules)

Entry Conditions:

- CNN/RNN divergence >50% for 10 consecutive minutes
- Model confidence oscillating wildly (std dev >0.3)

VFR Operations:

- Disable neural features (CNN/RNN)
- Trade on ILS regression + Order Flow only
- Reduce position sizes to 50%
- Tighten stops to 1× ATR

Exit VFR:

- CNN/RNN agreement >75% for 20 consecutive minutes
- AND ILS R² >0.85
- AND no circuit breakers active

Network Failure States

Primary Feed Down (WebSocket lag >3s):

- Switch to backup REST API
- Reduce update frequency to 5-second intervals
- Halt scalping strategies (M1)
- Log all trades during degraded mode

Both Feeds Down (>5s lag):

- Enter "Blind Mode"
- Close all positions at market
- Cancel all pending orders
- Halt system completely
- Alert operator via SMS/Telegram

Model Divergence Handling

CNN vs RNN Conflict:

- If disagreement >50% for 10 minutes → VFR mode
- If disagreement persists >1 hour → halt system
- Log state vectors for post-analysis

Live Performance Drift:

If $\text{Live_Sharpe_30d} < 0.7 \times \text{Backtest_Sharpe}$:

- Flag for model retraining
- Reduce position sizes to 50%
- Increase confidence threshold to 0.70

PHASE 6: MONITORING & OPTIMIZATION

Real-Time Dashboards

Primary Monitor:

- Current positions (symbol, size, P&L, time held)
- Confidence scores per symbol
- System state (EXECUTE/HOLD/ABORT/VFR)

- Circuit breaker status
- Network latency (real-time)

Secondary Monitor:

- Performance metrics (Sharpe, MaxDD, Win Rate)
- Resource utilization (CPU, RAM, ANE, Thermal)
- Order book depth visualization
- Recent trades log

Optimization Cycles

Frequency	Component	Method	Validation
1 minute	Pareto scoring	Rolling recalc	Top 20% selection
15 minutes	ILS parameters	Refit regression	R² >0.85 check
1 hour	Fuzzy weights	Grid search (last 50 trades)	Sharpe improvement
6 hours	HMM parameters	Baum-Welch algorithm	State transition accuracy
24 hours	Slippage model	Actual fill analysis	Error <10%
1 week	CNN/RNN	Retrain on recent data	Out-of-sample Sharpe >1.0

Performance Logging

Every Trade Record:

```
{
  trade_id: UUID
  timestamp_entry: ISO8601
  timestamp_exit: ISO8601
  symbol: string
  direction: LONG/SHORT
  entry_price: float
  exit_price: float
  size: float
  confidence_at_entry: float
  model_trigger: CNN/RNN/BOTH/ILS
  hmm_regime: TREND/RANGE
  predicted_slippage: float
  actual_slippage: float
  pnl: float
  pnl_percentage: float
  exit_reason: string
  notes: string
}
```

Daily Aggregates:

- Total trades
 - Win rate
 - Avg win / Avg loss
 - Sharpe ratio
 - Max drawdown
 - Effective edge distribution
 - Slippage accuracy
-

PHASE 7: HARDWARE OPTIMIZATION (Mac Studio M2)

Thread Allocation

Thread 1 (Core ML):

- Data ingestion from WebSocket
- Fibonacci normalization
- Priority: Real-time (highest)

Thread 2 (ANE - Apple Neural Engine):

- CNN inference (pattern recognition)
- Batch size: 5 symbols parallel
- Priority: High

Thread 3 (ANE):

- RNN inference (trend memory)
- Batch size: 5 symbols parallel
- Priority: High

Thread 4 (Performance Cores):

- Brute force optimization
- HMM regime detection
- KNN strategy selection
- Priority: Medium

Thread 5 (Efficiency Cores):

- Risk monitoring
- Circuit breaker checks
- Performance logging
- Priority: Medium-Low

Thermal Management

Temperature Monitoring:

- Check ANE temperature every 30 seconds
- If $>80^{\circ}\text{C}$: increase inference interval from 1min $\rightarrow$ 5min
- If $>85^{\circ}\text{C}$: halt neural inference, switch to ILS-only mode
- Log all thermal throttling events

Memory Management

RAM Allocation:

- Live data buffer: 4GB (last 24h OHLCV)
- Model weights: 2GB (CNN/RNN)
- State history: 1GB (latent vectors)

- Logs and monitoring: 1GB
- System overhead: 2GB
- Reserve: 22GB (for Mac OS and other processes)

Cleanup Routines:

- Purge raw OHLCV >60 minutes old (keep latent vectors only)
 - Compress logs >24 hours old
 - Archive completed trades to disk
-

PHASE 8: PRE-LIVE VALIDATION

Backtest Requirements

Data:

- 2 years historical data minimum
- Include at least 2 bear markets and 2 bull markets
- Verify data quality (no gaps, outliers flagged)

Metrics to Achieve:

- Sharpe Ratio >1.5
- Max Drawdown <15%
- Win Rate >55%
- Profit Factor >1.8
- Effective Edge >0.15 on 80%+ trades

Walk-Forward Analysis

Segments:

1. Train: Jan-Jun 2024
2. Validate: Jul-Sep 2024
3. Out-of-Sample: Oct-Dec 2024
4. Live Simulation: Jan-Feb 2025

Requirements:

- Out-of-sample Sharpe within 30% of in-sample

- Max drawdown not exceeding 2× backtest
- No parameter curve-fitting detected

Monte Carlo Stress Testing

- Run 1,000 random path simulations
- Test worst backtest day occurring 3× consecutively
- Verify Kelly fractions never exceed 0.25
- Confirm circuit breakers trigger correctly

Paper Trading Phase

Duration: 30 days minimum

Goals:

- Achieve Sharpe >1.2 (allowing for live friction)
- Max drawdown <15%
- Effective edge >0.15 on 80%+ of trades
- Zero execution errors
- Latency <200ms (P99)

Validation Checklist:

- ☐ All order types execute correctly
- ☐ Circuit breakers trigger as designed
- ☐ VFR mode activates/deactivates properly
- ☐ Network failover tested
- ☐ Manual kill switch works (<2s full stop)
- ☐ Slippage model accuracy within 10%
- ☐ Live Sharpe within 20% of backtest

PHASE 9: GO-LIVE DEPLOYMENT

Capital Allocation Schedule

Day	Capital %	Allowed Strategies	Success Metric
0-7	5%	M15+ only, max 2 positions	Zero execution errors
8-14	5%	M15+, max 3 positions	Live Sharpe >1.0

Day	Capital %	Allowed Strategies	Success Metric
15-21	10%	M5+, max 4 positions	Effective edge >0.15
22-30	10%	M5+, max 5 positions	Live Sharpe >1.2
30+	Full	All timeframes	Metrics within 20% of backtest

Pre-Flight Checklist (Daily)

Before Market Open:

- ☐ Network latency check (P99 <200ms)
- ☐ API keys active and rate limits confirmed
- ☐ Backup internet connection tested
- ☐ UPS battery charge >90%
- ☐ Disk space >500GB free
- ☐ Mac temperature <60°C
- ☐ All monitoring dashboards operational
- ☐ Manual kill switch tested

Kill Switch Protocol

Keyboard Shortcut: Cmd+Shift+K

Execution Sequence (auto-timed):

1. Cancel all pending orders (0.5s)
2. Close all positions at market (2.0s)
3. Dump system state to JSON (3.0s)
4. Halt all threads (4.0s)
5. Alert operator (5.0s)

Test monthly during off-hours

PHASE 10: OPERATIONAL PROCEDURES

Daily Routine

08:00 Bangkok Time:

- Review overnight performance
- Check for system alerts/warnings

- Verify API connectivity to all exchanges
- Review top Pareto-selected symbols

12:00 Bangkok Time:

- Mid-day performance check
- Verify no circuit breakers triggered
- Check thermal status
- Review confidence score distribution

20:00 Bangkok Time:

- End-of-day P&L reconciliation
- Export trade log
- Check for model drift
- Plan symbol rotation for next day

23:00 Bangkok Time:

- Full system backup
- Log rotation and compression
- Verify tomorrow's market schedule (holidays, halvings, etc.)

Weekly Routine

Sunday Evening:

- Full system restart (clear RAM)
- Model retraining (if drift detected)
- Update KNN database with week's successful trades
- Recalibrate slippage model with actual fills
- Review Sharpe by symbol, model, and regime
- Adjust Pareto threshold if needed

Monthly Routine

First Sunday:

- Full backtest on last month's data
- Compare live vs backtest performance
- Update HMM training dataset

- Test manual kill switch
- Review and archive logs >30 days
- Hardware health check (SSD, RAM, thermal paste)

Incident Response

Performance Degradation (Sharpe drops >50%):

1. Reduce position sizes to 50%
2. Increase confidence threshold to 0.75
3. Enable enhanced logging
4. Conduct model drift analysis
5. If issue persists >3 days, halt for full audit

Exchange API Issues:

1. Switch to backup exchange (if available)
2. If unavailable, close positions and go flat
3. Monitor exchange status pages
4. Resume when confirmed stable for 1 hour

Hardware Failure:

1. Activate kill switch immediately
2. Switch to backup Mac Mini (if configured)
3. Restore from latest backup
4. Verify state integrity before resuming

SYSTEM CONFIGURATION FILES

config/global.yaml

```
System:
name: "Master Flight Controller"
version: "2026.1"
environment: "production" # production | paper | backtest

Hardware:
device: "mac_studio_m2"
neural_engine: true
```

thread_count: 5

Capital:

total: 100000 # USD

max_deployment: 0.50 # 50% max

reserve: 0.50 # 50% cash reserve

Latency:

bangkok_tokyo_p99: 150 # milliseconds

bangkok_london_p99: 280

bangkok_newyork_p99: 320

update_frequency: 900 # 15 minutes

config/layers.yaml

Layer0_OrderFlow:

enabled: true

weight: 0.05

normalization: "tanh"

Layer1_Fibonacci:

lookback_hours: 24

update_minutes: 15

levels: [0, 0.236, 0.382, 0.5, 0.618, 0.786, 1.0]

Layer2_Neural:

cnn:

input_minutes: 15

update_minutes: 1

device: "ane"

weight: 0.25

rnn:

input_minutes: 60

update_minutes: 1

device: "ane"

weight: 0.25

Layer3_Dispatcher:

hmm:

states: ["trend", "range"]

update_minutes: 15

retrain_hours: 6

weight: 0.10

pareto:

```
percentile: 0.20 # top 20%
update_minutes: 5
weight: 0.15
correlation:
  window_minutes: 30
  threshold: 0.85
```

Layer4_ILS:

```
regression_points: 60
refit_minutes: 15
r_squared_threshold: 0.85
weight: 0.20
```

Layer5_Fuzzy:

```
zones:
  execute: 0.85
  scale_in: 0.65
  hold: 0.45
  scale_out: 0.30
  abort: 0.00
hysteresis_minutes: 5
hysteresis_buffer: 0.05
```

config/risk.yaml

Position_Sizing:

```
method: "pareto_kelly"
max_kelly_fraction: 0.25
lookback_trades: 50
```

Hard_Limits:

```
max_per_symbol: 0.30 # 30%
max_concurrent: 5
max_total_deployment: 0.50 # 50%
```

Circuit_Breakers:

level1:

```
drawdown: 0.05 # 5%
window_minutes: 60
action: "halt_entries_2h"
```

level2:

```
drawdown: 0.10 # 10%
window_hours: 24
action: "reduce_positions_50pct"
```

```
level3:
  price_move: 0.05 # 5% in 30s
  action: "emergency_mode"
```

```
Stop_Loss:
  normal: 2.0 # x ATR
  turbulent: 1.0
  unstable: 0.5
```

config/execution.yaml

```
Friction_Layer:
  effective_edge_minimum: 0.15
  slippage_model:
    base_spread: "exchange_specific"
    depth_weight: 1.0
    volatility_weight: 1.0
    latency_multiplier: 2.0
```

```
Order_Routing:
  aggressive_limit:
    confidence_min: 0.85
    ticks_inside: 1
    timeout_seconds: 2
    fallback: "market"
  passive_limit:
    confidence_min: 0.65
    price: "mid"
    timeout_seconds: 10
    fallback: "cancel"
  market:
    confidence_max: 0.30
    action: "immediate"
```

```
Exit_Logic:
  urgent:
    confidence: 0.30
    method: "market"
  strategic:
    confidence: 0.45
    method: "twap_60s"
  partial:
    drop: 0.20
    window_minutes: 5
```

BACKUP & DISASTER RECOVERY

Backup Strategy

Real-Time (Continuous):

- Trade log: append-only database
- State vectors: write every 15 minutes
- Model weights: version control

Daily (00:00 Bangkok Time):

- Full system state snapshot
- Configuration files
- Performance metrics
- Logs (compressed)

Weekly (Sunday 23:00):

- Historical data archive
- Model training datasets
- KNN database
- Full disk image

Storage Locations:

- Primary: Local SSD (encrypted)
- Secondary: External SSD (daily sync)
- Tertiary: Cloud storage (weekly archive)

Recovery Procedures

Minor Failure (Single Thread Crash):

- Auto-restart affected thread
- Reload last known good state
- Resume operations within 30 seconds

Major Failure (System Crash):

1. Boot from backup disk image
2. Restore last state snapshot
3. Verify data integrity
4. Run system diagnostics
5. Resume in VFR mode for 1 hour
6. Switch to full auto if stable

Data Corruption:

1. Halt system immediately
2. Restore from last verified backup
3. Replay trade log to recover state
4. Validate P&L reconciliation
5. Manual approval before resuming

Failover Configuration

Backup Hardware (Optional Mac Mini M2):

- Identical software configuration
 - Receives state snapshots every 15 minutes
 - Auto-activates if primary goes offline >5 minutes
 - Runs in read-only monitoring mode normally
-

MAINTENANCE SCHEDULE

Daily

- Verify network connectivity
- Check thermal status
- Review overnight trades
- Monitor circuit breaker triggers

Weekly

- Restart system (RAM flush)
- Update slippage model

- Review model performance by regime
- Backup rotation

Monthly

- Full backtest on recent data
- Hardware health check
- Test kill switch
- Review and optimize fuzzy weights
- Archive old logs

Quarterly

- CNN/RNN retraining on 6-month dataset
- HMM parameter full recalibration
- Performance audit vs initial backtest
- Hardware upgrade evaluation

APPENDIX A: SYMBOL SELECTION CRITERIA

Crypto:

- BTC/USDT, ETH/USDT (always included)
- Market cap >\$5B
- 24h volume >\$500M
- Listed on 3+ major exchanges

Forex:

- Major pairs only (EUR/USD, GBP/USD, USD/JPY, etc.)
- Avoid exotic pairs (high spread)
- Consider Bangkok session overlap

Commodities:

- Gold, Silver, Oil (WTI, Brent)
- High liquidity futures only

Indices:

- S&P 500, NASDAQ, DAX
 - Via CFD or futures
-

APPENDIX B: EXCHANGE API REQUIREMENTS

Minimum Features:

- WebSocket OHLCV streaming
- WebSocket order book (Level 2)
- REST order placement (limit, market, stop)
- REST order cancellation
- REST position query
- REST balance query

Rate Limits:

- WebSocket: unlimited (connection-based)
- REST: minimum 100 requests/minute
- Order placement: minimum 10 orders/second

Latency SLA:

- WebSocket: <100ms from exchange to client
 - REST: <200ms round-trip
-

APPENDIX C: ALERT & NOTIFICATION SYSTEM

Critical Alerts (SMS + Telegram):

- Circuit breaker Level 2 or 3 triggered
- Network down >5 minutes
- Hardware temperature >85°C
- Manual intervention required

Warning Alerts (Telegram only):

- Circuit breaker Level 1 triggered
- VFR mode activated

- Model drift detected
- Latency >200ms for >1 hour

Info Notifications (Dashboard only):

- Daily performance summary
 - Weekly optimization results
 - New symbol selected by Pareto filter
-

APPENDIX D: GLOSSARY

ANE: Apple Neural Engine - hardware accelerator for ML inference **ATR:** Average True Range - volatility measure **HMM:** Hidden Markov Model - regime detection **ILS:** Instrument Landing System - regression-based guidance **KNN:** K-Nearest Neighbors - pattern matching **P99:** 99th percentile latency **Pareto:** 80/20 principle - focus on vital few **TWAP:** Time-Weighted Average Price - gentle exit method **VFR:** Visual Flight Rules - degraded mode without neural models

Installation Complete - System Ready for Validation Phase